Emre Tiryaki

Backend Developer



Samsun, Turkey
+90 544 319 52 15
tiryakiemre18@gmail.com
emre-tiryaki.dev
in/emre-tiryaki
emre-tiryaki

Profile Summary

Software engineering student focused on distributed systems, microservice architectures, and developing scalable backend applications. Builds applications with Go, Java, TypeScript, and Rust; designs modular, testable, and maintainable services using NestJS, Spring Boot, Gin, GraphQL, and RabbitMQ. Adopts modern software development practices by actively utilizing Docker, Linux, and CI/CD pipelines.

Education

Inonu University — Malatya, Turkey
Software Engineering (B.Sc.) — *2023 – 2027 (Expected)*

- 4th-year student; focus areas: backend development, DevOps, distributed systems.

Technical Skills

Programming Languages	TypeScript, JavaScript, Go, Rust
Backend	Gin, Express, Axum, Fastify, NestJS, GraphQL, REST, RabbitMQ
Databases	PostgreSQL, MySQL, MongoDB
Distributed Systems	Asynchronous messaging, inter-service communication, scalable architecture design
DevOps / Infrastructure	Docker, CI/CD, Linux, Git

Experience

Avikon – Backend Developer Intern — August 2026 – September 2026

- Developed and maintained Fastify-based backend services.
- Designed and implemented a GraphQL API layer using the Mercurius library.
- Built an asynchronous inter-service messaging infrastructure utilizing RabbitMQ.

Projects

Autonomous Financial Assistant (OFA) – Smart Payment Routing and Problem Solver — July 2026
Moka United AI Ideathon & Hackathon — JavaScript, Express, Python, Machine Learning, React

- Developed a machine learning-based algorithm to select the most optimal acquirer bank for transactions.
- Designed an AI agent architecture that takes autonomous actions to resolve system issues without requiring human intervention.

OKUR – AI Exam Grader — May 2026
Inufest 2026 Competition — TypeScript, NestJS, Python, Machine Learning, React

- Built a system that automatically generates exam papers in LaTeX format.
- Implemented an image processing module that crops and transcribes student answers from scanned physical papers.
- Developed an AI model that semantically compares student answers with the instructor's answer key to automate grading.

Cosmic Data Processing Pipeline – Big Data Processing Engine — March 2026
TUA Astro Hackathon — Python, FastAPI, React

- Built a component to generate synthetic cosmic data and transmit it directly to the processing engine.
- Implemented an end-to-end data processing pipeline utilizing real-time digital signal processing (DSP) via NumPy and SciPy vectorization to clean cosmic radiation-induced signal degradation.
- Achieved 90%+ autonomous anomaly detection accuracy using LSTM Autoencoder models.

HEGS News – Gamified News Platform — October 2025
Anadolu Agency Media Technologies Hackathon — JavaScript, Java, Spring Boot, Python, FastAPI

- Scraped news data from the Anadolu Agency website and presented it through a clean, user-friendly interface.
- Integrated an AI module to provide users with dynamically generated summaries of current news.

Languages

Turkish Native
English Professional working proficiency

Independent Work

- Enhanced Bash scripting skills on Arch Linux, utilizing it as a daily driver operating system.
- Developed `project-toolkit`, a bash script automating project creation and configuration workflows.
- Created `auto-readme-generator` (Rust/Axum backend, React/JavaScript frontend), a tool that scans repositories to automatically generate README files using user-specified AI models.
- Built `pdf2markdown`, a Python CLI tool that converts PDF files to Markdown concurrently via locally hosted Ollama models.
- Developed `RepoGrab`, a Go-based TUI (Terminal User Interface) tool allowing users to download individual files from GitHub and GitLab without cloning the entire repository.